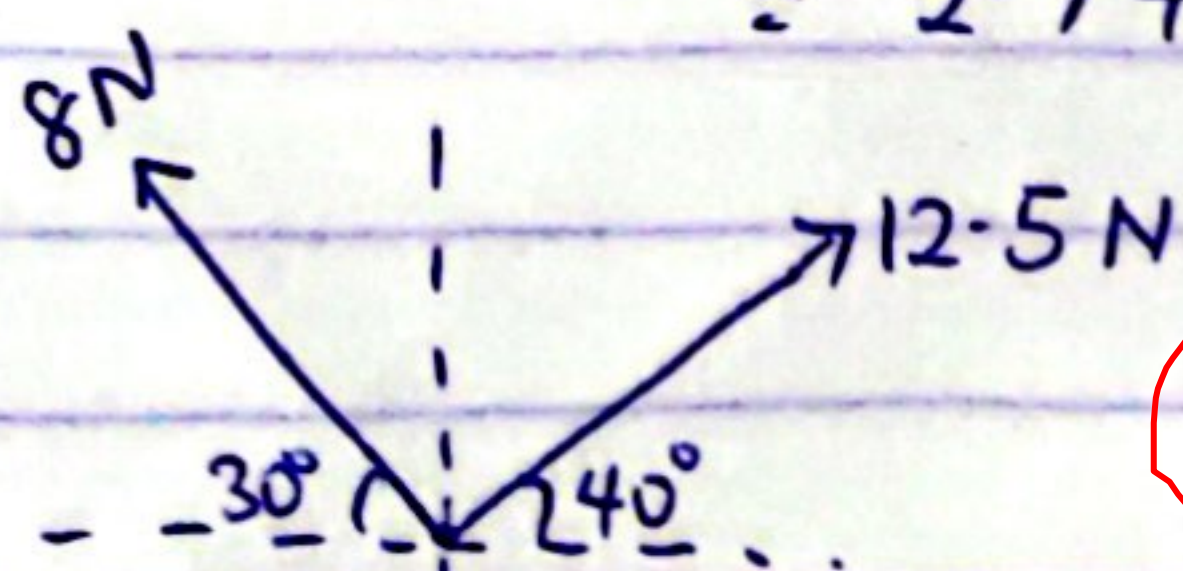


S-6 PHYSICS 2026 TEST 2 GUIDE

(a) Frictional force, $F = \mu R$
 $= 0.4 \times 0.7 \times 9.81$
 $= 2.7468 \text{ N}$ ✓



$$F_x = 12.5 \cos 40 - 8 \cos 30$$

$$= 2.6474 \text{ N}$$
 ✓

$$F_y = 12.5 \sin 40 + 8 \sin 30$$

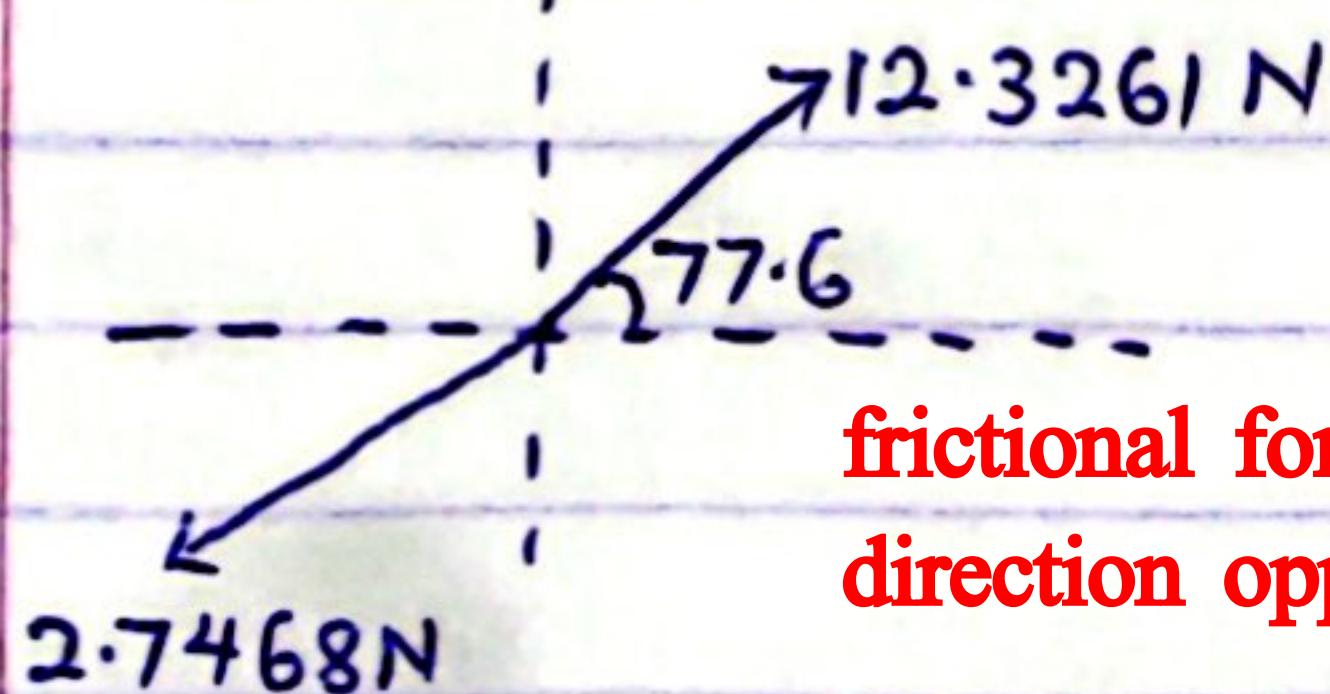
$$= 12.0384 \text{ N}$$
 ✓

$$\tan \alpha = \frac{12.0384}{2.6474}$$

$$\alpha = 77.6^\circ$$
 ✓

$$F = \sqrt{2.6474^2 + 12.0384^2}$$

$$= 12.3261 \text{ N}$$
 ✓



frictional force acts in a direction opposite to motion

$$\text{Resultant force} = 12.3261 - 2.7468$$

$$= 9.5793 \text{ N}$$

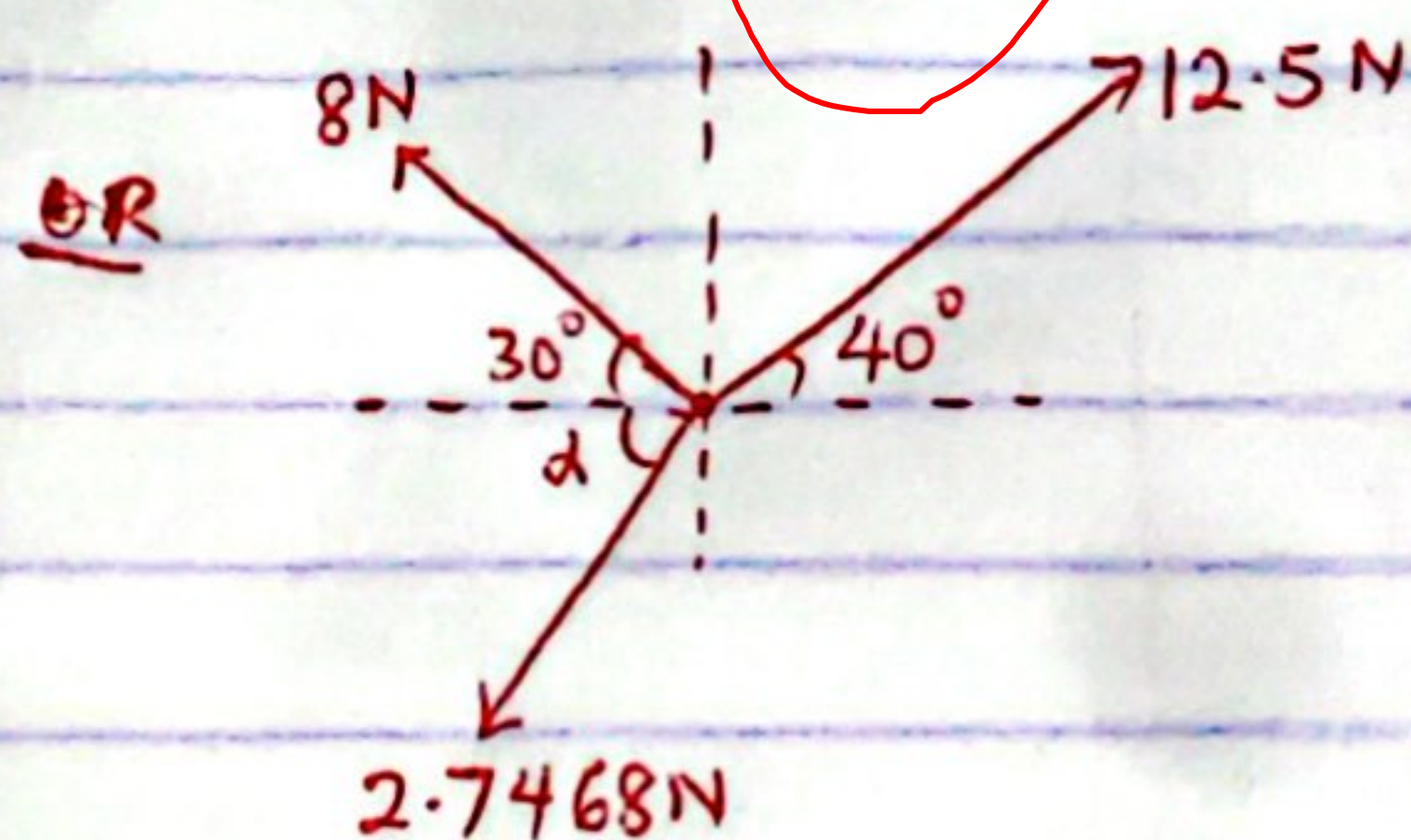
$$F = ma$$
 ✓

$$9.5793 = 0.7 \times a$$

$$a = 13.6847 \text{ m/s}^2$$
 ✓

∴ The robot moves with an acceleration of 13.6847 m/s^2 at an angle of 77.6° to the horizontal

26
26



$$R_x = R \cos \alpha = 12.5 \cos 40 - 8 \cos 30 - 2.7468 \cos \alpha$$

$$(R + 2.7468) \cos \alpha = 2.6474 \quad \text{--- (1)}$$

$$R_y = R \sin \alpha = 12.5 \sin 40 + 8 \sin 30 - 2.7468 \sin \alpha$$

$$(R + 2.7468) \sin \alpha = 12.0384 \quad \text{--- (2)}$$

$$\text{(1)}^2 + \text{(2)}^2$$

$$(R + 2.7468)^2 = 2.6474^2 + 12.0384^2$$

$$R = 9.5793 \text{ N}$$

$$\text{(2)} \div \text{(1)}$$

$$\tan \alpha = \frac{12.0384}{2.6474}$$

$$\alpha = 77.6^\circ$$

$$F = ma$$

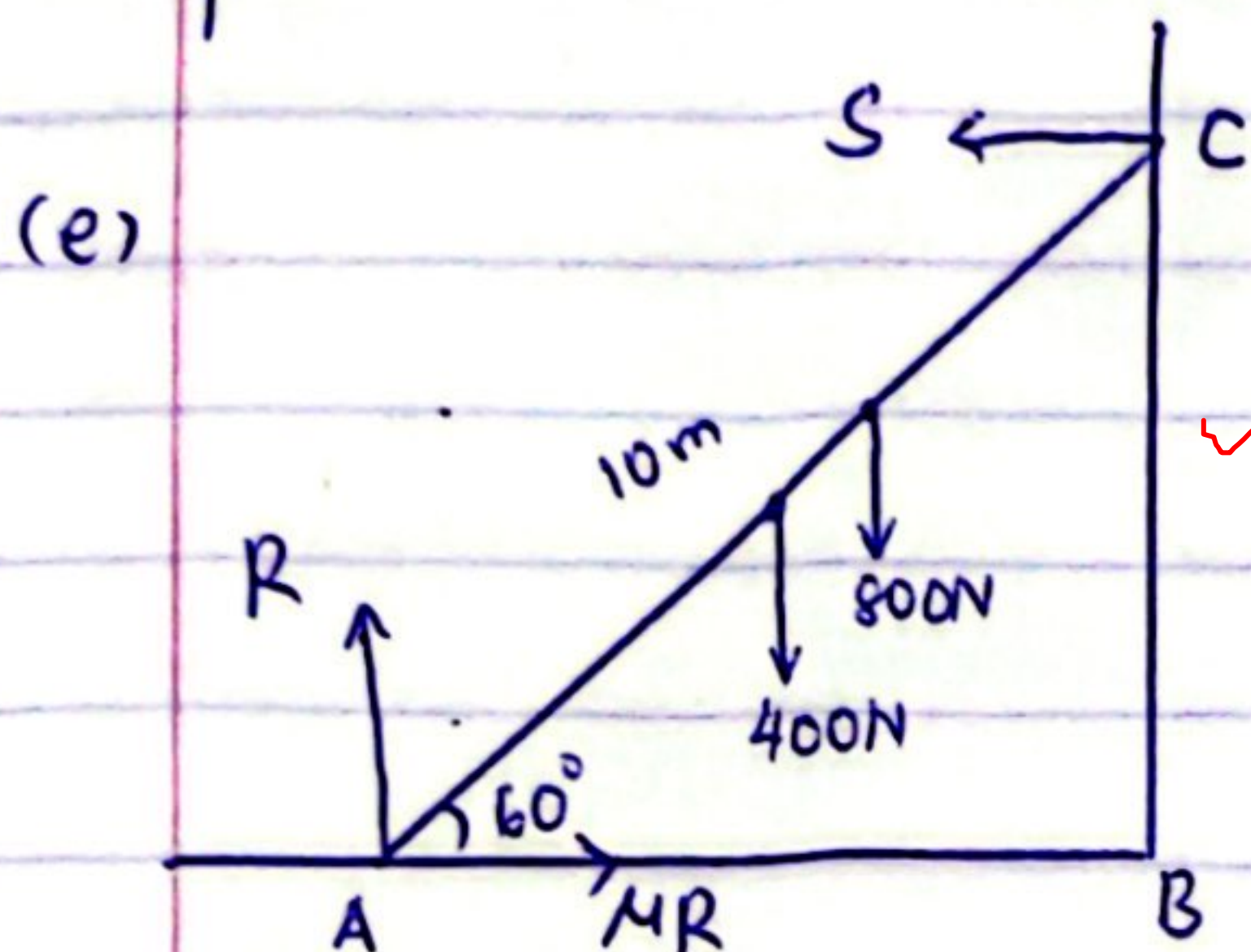
$$9.5793 = 0.70 a$$

$$a = 13.6847 \text{ m/s}^2$$

(b) Long spanner. The length is increased, this increases the moment of the arm when a small force is applied causing a large turning effect.

(c) A couple is a pair of equal, parallel and opposite forces with different lines of action. ✓
A couple rotates the jar lid. ✓

- (d) The forces act on the ladder in a way that:
1. The algebraic sum of all the forces on the ladder in any direction is zero i.e. no movement. ✓
 2. The algebraic sum of moments of all the forces about any point is zero i.e. no rotation. ✓



01- illustration

$$S = MR \quad \text{--- ①}$$

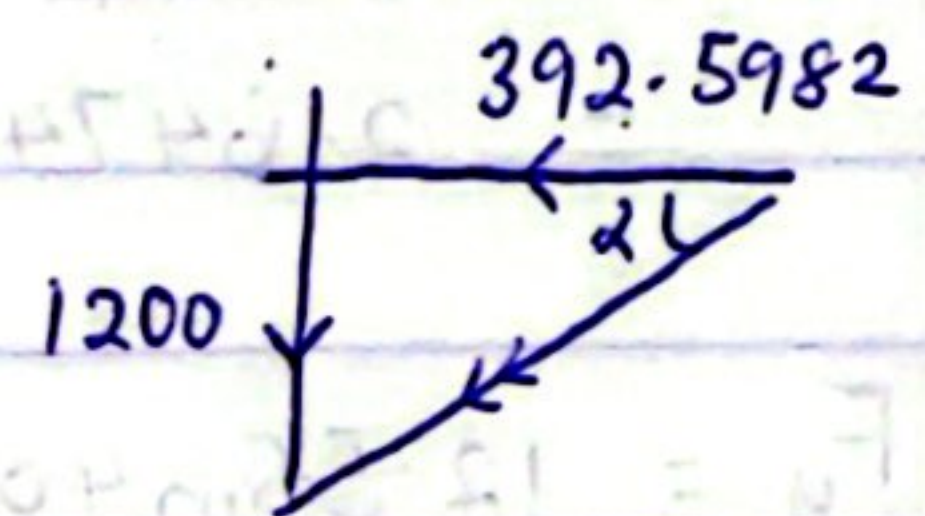
$$R = 400 + 800$$

$$R = 1200 \text{ N} \quad \text{--- ②}$$

Taking moments about A;

$$S \times 10 \sin 60 = 400 \times 5 \cos 60 + 800 \times 6 \cos 60$$

$$S = 392.5982 \text{ N}$$



Reaction on the

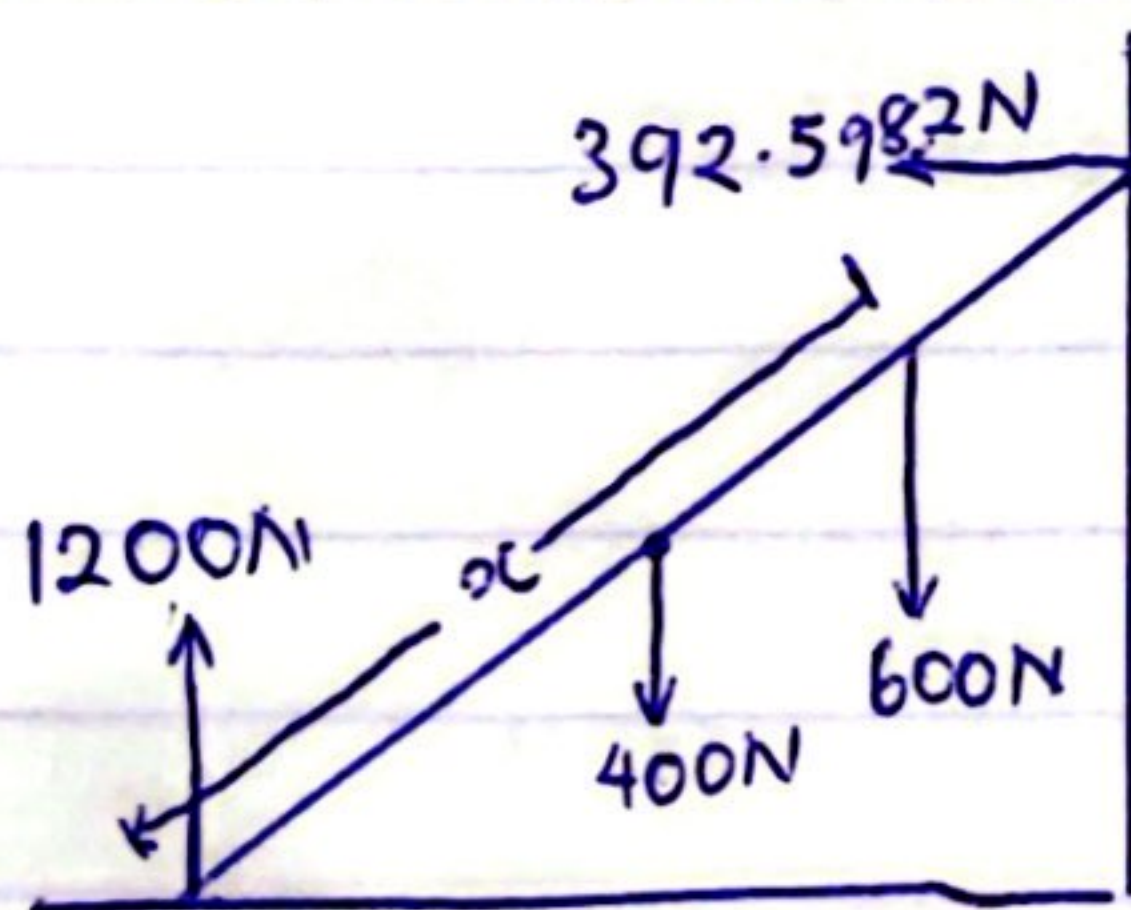
$$\text{Ground} = \sqrt{1200^2 + 392.5982^2}$$

$$= 1262.5899 \text{ N}$$

$$\tan \alpha = \frac{1200}{392.5982}$$

$$\alpha = 71.88^\circ$$

(f)



Taking moments about A;

$$392.5982 \times 10 \sin 60 = 400 \times 5 \cos 60 + 600 \times x \cos 60$$

$$x = 8.0000 \text{ m}$$

∴ The technician should climb up the ladder a distance of 8.0000m

END